Why Cross-Method Circuit Consensus Fails: An Empirical Analysis

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Abstract

Circuit discovery methods—such as Edge Attribution Patching (EAP-IG), Edge Pruning (EP), and Automated Circuit Discovery (ACDC)—aim to identify minimal subgraphs that faithfully explain model behavior. A natural hypothesis is that *consensus* across independently derived circuits should yield higher-quality, more faithful subgraphs. We systematically evaluate this hypothesis through *Cross-method Structural consensus Scoring* (CSS), a binary consensus framework applied to $K \geq 3$ circuit discovery methods across multiple tasks and models. Our findings are predominantly negative. On GPT-2 Small, CSS produces a faithful consensus circuit only on Indirect Object Identification (IOI), while failing on Subject–Verb Agreement (SVA) and Arithmetic—a 1/3 success rate. Moreover, the sole positive result is confounded by two evaluation artifacts: ACDC's `isinf` binarization and zero-ablation size bias. On Qwen-2.5 0.5B (179k edges), all methods exhibit near-zero pairwise overlap (Jaccard ≈ 0.04), rendering consensus trivially empty. We identify three structural causes of failure: (1) high inter-method correlation on most tasks eliminates the independent-evidence premise of consensus; (2) ACDC binarization pathologies inflate apparent method diversity; and (3) combinatorial dilution in larger edge spaces systematically destroys cross-method overlap. These results suggest that binary circuit consensus is viable only within a narrow "Goldilocks zone" of inter-method agreement, which rarely obtains in practice.

1 Introduction

Mechanistic interpretability aims to understand neural network computation by identifying *circuits*—subgraphs of the computational graph that are necessary and sufficient for a given behavior [Olah et al., 2020, Wang et al., 2023, Conmy et al., 2023]. Multiple circuit discovery methods have been proposed, including gradient-based attribution [Syed et al., 2023], edge pruning [Hanna et al., 2024], and automated search [Conmy et al., 2023]. A persistent challenge is that different methods often identify substantially different circuits for the same task [Mondorf and Plank, 2025], raising the question: which edges are "real"?

One intuitive approach is *consensus*: edges identified by multiple independent methods may constitute a more reliable circuit. This idea has been explored through score averaging [Mondorf and Plank, 2025] and within-method threshold diversity [CIRCUS; Parekh et al., 2026], but systematic cross-method binary consensus remains understudied.

We introduce *Cross-method Structural consensus Scoring* (CSS), which assigns each edge a score $s \in \{1/K, 2/K, \ldots, 1\}$ based on the fraction of K methods that include it. The CSS$= 1.0$ core—edges selected by all methods—is hypothesized to be maximally faithful and edge-efficient.

Our extensive experiments reveal that this hypothesis **fails in the majority of settings**. Across three tasks on GPT-2 Small and two tasks on Qwen-2.5 0.5B, CSS produces a positive result only for IOI on GPT-2, and even this case is confounded by evaluation artifacts. Rather than presenting CSS as a method, we pivot to analyzing *why* consensus fails, yielding three contributions:

1. **Method correlation undermines consensus.** On SVA and Arithmetic, gradient-based methods (EAP-IG, EP) exhibit high pairwise Jaccard (≥ 0.65), violating the independent-evidence as-

sumption. The resulting consensus adds no information beyond the most selective single method (§3).

2. **Binarization pathologies inflate diversity.** ACDC's default `isinf` threshold produces extremely sparse circuits (14–17 edges vs. 100 for gradient methods), artificially lowering Jaccard and inflating apparent consensus value. Top-k normalization partially corrects this but does not rescue consensus on failing tasks (§4).

3. **Combinatorial dilution destroys consensus at scale.** On Qwen-2.5 0.5B (179k edges), the same top-0.3% binarization that works on GPT-2 (32k edges) yields near-zero Jaccard (≤ 0.042) and an empty CSS$= 1.0$ set, demonstrating that edge-level consensus does not scale (§5).

2 Background and Method

2.1 Circuit Discovery Methods

We employ three methods from different algorithmic families:

EAP-IG (Edge Attribution Patching with Integrated Gradients) computes edge importance via the integrated gradient of the patching effect along an interpolation path between clean and corrupted activations [Syed et al., 2023].

EP (Edge Pruning) learns a continuous mask over edges, optimizing for faithfulness while encouraging sparsity through L_1 regularization [Hanna et al., 2024].

ACDC (Automated Circuit Discovery) iteratively prunes edges by testing whether each edge's removal changes model output beyond a threshold [Conmy et al., 2023].

2.2 Cross-method Structural Consensus Scoring

Given K circuit discovery methods, each producing a binary edge set $C_k \subseteq E$ over the full edge set E of a transformer, we define:

$$\text{CSS}(e) = \frac{1}{K} \sum_{k=1}^{K} \mathbf{1}[e \in C_k] \tag{1}$$

The CSS$= 1.0$ core is $\{e : \text{CSS}(e) = 1\}$, containing edges selected by all K methods.

2.3 Evaluation Metrics

We evaluate circuits using:

- **Circuit Performance Recovery (CPR)**: the fraction of clean-model performance recovered by the circuit. CPR $= 1.0$ means perfect recovery.
- **Circuit Minimality Deviation (CMD)**: a measure of circuit size relative to the full graph. Lower is better.
- **Pairwise Jaccard**: $J(C_i, C_j) = |C_i \cap C_j|/|C_i \cup C_j|$, measuring overlap between two method outputs.

Seed	CSS$=1.0$ CPR	ACDC CPR	Δ
42 (original)	1.187	1.175	$+0.013$
123	1.168	1.179	-0.011
456	1.178	1.182	-0.004
789	1.175	1.170	$+0.005$
1024	1.171	1.174	-0.003
Mean $\pm$ std	1.176 ± 0.007	1.176 ± 0.005	0.000 ± 0.009

Table 1: SVA multi-seed bootstrap analysis. The apparent advantage at seed 42 vanishes under replication; the 95% CI for ΔCPR includes zero.

Condition	Result	Note
$J_{\max} < 0.70$	PASS (0.653)	marginal
$\lvert\text{CSS} \geq 2/3\rvert \geq 10$	PASS (54)	
$\lvert\text{CSS} = 1.0\rvert \geq 3$	PASS (44)	
CPR dominance	FAIL ($\Delta \approx 0$)	CI includes zero

Table 2: SVA diagnostic conditions. The high inter-method Jaccard (0.653) and absence of CPR dominance indicate consensus failure.

2.4 Binarization

A critical preprocessing step is converting continuous scores (EAP-IG, EP) or variable-size outputs (ACDC) into comparable binary edge sets. We use **top-k** binarization: select the top k edges by importance score for each method. We also examine ACDC's default **`isinf`** binarization, which retains only edges with infinite importance (those whose removal changes the output token), typically yielding 14–17 edges.

3 GPT-2 Small Experiments

We evaluate CSS with $K = 3$ (EAP-IG, EP, ACDC) on three tasks using GPT-2 Small (117M parameters, 32,491 total edges).

3.1 Indirect Object Identification (IOI)

IOI requires the model to predict the indirect object in sentences like "When Mary and John went to the store, John gave a drink to _____".

Results. With top-k binarization ($k = 100$), pairwise Jaccard values are moderate: $J(\text{EAP-IG}, \text{EP}) = 0.307$, $J(\text{EAP-IG}, \text{ACDC}) = 0.093$, $J(\text{EP}, \text{ACDC}) = 0.133$. The CSS$= 1.0$ core contains 7 edges with CPR $= 0.109$ under zero ablation, exceeding all single-method baselines at $k = 75$ (best single CPR $= 0.029$). This is the **only positive result** across all experiments.

3.2 Subject–Verb Agreement (SVA)

SVA requires predicting the correct verb number in sentences with intervening attractors.

Results. The gradient methods are highly correlated: $J(\text{EAP-IG}, \text{EP}) = 0.653$, near the failure threshold. The CSS$= 1.0$ core contains 44 edges with mean CPR $= 1.173$, while the best single method (ACDC at $k = 75$) achieves CPR $= 1.175$. Multi-seed bootstrap analysis ($n = 5$ seeds) yields $\Delta\text{CPR} = 0.000 \pm 0.009$ with 95% CI $[-0.012, +0.019]$ spanning zero. Table 1 shows per-seed results.

Task	$J_{\max}$	CSS$=1.0$ CPR	Best Single	Verdict
IOI	0.307	$+0.109$	$+0.029$	PASS
SVA	0.653	$+1.173$	$+1.175$	FAIL
Arithmetic	0.695	$+0.919$	$+0.985$	FAIL

Table 3: GPT-2 Small results across three tasks. CSS shows advantage only on IOI, where inter-method Jaccard is lowest. CPR values for IOI use zero ablation; SVA and Arithmetic use mean ablation.

Verdict: FAIL. The high $J(\text{EAP-IG}, \text{EP})$ means gradient methods dominate the consensus, and CSS reduces to a noisy version of ACDC.

3.3 Arithmetic (Two-Digit Addition)

The model must produce the correct sum for prompts like "47 + 38 =".

Results. Inter-method correlation is the highest across all tasks: $J(\text{EAP-IG}, \text{EP}) = 0.695$. Under mean ablation, CSS$= 1.0$ achieves CPR $= 0.919$ vs. best single method CPR $= 0.985$, a *deficit* of -0.066.

Verdict: FAIL. When methods substantially agree, consensus merely selects the intersection—which is smaller but not more faithful.

3.4 Factual Recall (Abandoned)

We attempted factual recall (country–capital pairs) on GPT-2 Small but abandoned the experiment due to task–model mismatch: baseline accuracy was only 52% ($\pm 15\%$), barely above the 25% random baseline. ACDC could not converge under such unstable baselines, and gradient methods produced circuits with near-zero faithfulness (EAP-IG $k = 50$: CPR $= 0.02$; EP $k = 50$: CPR $= -0.01$). GPT-2 Small lacks sufficient parametric knowledge for localized factual retrieval circuits.

3.5 GPT-2 Summary

Table 3 summarizes all GPT-2 results. CSS succeeds only on IOI (1/3 tasks), where max pairwise Jaccard is lowest (0.307).

4 Binarization and Ablation Confounds

The sole positive result (IOI) is subject to two confounds that we analyze in detail.

4.1 ACDC `isinf` vs. Top-k Binarization

ACDC's default `isinf` threshold retains only 14 edges on IOI, compared to 100 for gradient methods at $k = 100$. This extreme sparsity mismatch artificially deflates $J(\text{ACDC}, \cdot)$ and makes CSS$= 1.0 \approx$ ACDC.

Top-k normalization raises IOI's $J(\text{ACDC}, \cdot)$ from 0.09 to 0.25 (Table 4), but the CSS positive result persists—indicating that while `isinf` inflates apparent diversity, it is not the sole driver.

4.2 Top-k Sensitivity Sweep

We sweep $k \in \{50, 75, 100, 150\}$ across all GPT-2 tasks (Table 5).

CSS CPR is stable across k (std < 0.02 within each task), confirming that binarization threshold is not responsible for consensus failure. The failure is structural: method correlation, not k-sensitivity.

Task	J (isinf)	J (top-k)	ΔJ	Outcome change?
IOI	0.09	0.25	$+0.16$	No (still PASS)
Arithmetic	0.17	0.42	$+0.25$	No (still FAIL)
SVA	0.37	0.37	0	N/A (already top-k)

Table 4: Effect of ACDC binarization on Jaccard. `isinf` systematically deflates Jaccard by producing extremely sparse circuits.

Task	k=50	k=75	k=100	k=150	Stable?	Beats single?
IOI (zero)	0.109	0.102	0.095	0.088	Yes	Yes (all k)
SVA (mean)	1.185	1.187	1.174	1.168	Yes	No (all k)
Arith (mean)	0.919	0.919	0.926	0.933	Yes	No (all k)

Table 5: CSS$=1.0$ CPR across top-k values. CSS is stable across k (std < 0.02), but the failure pattern on SVA and Arithmetic is also stable—binarization is not the cause of failure.

4.3 Ablation Method Sensitivity

We compare zero, mean, and resample ablation on all three GPT-2 tasks (Table 6).

IOI's CSS advantage drops from $+0.080$ under zero ablation to $+0.007$ (mean) and $+0.003$ (resample). Zero ablation disproportionately penalizes larger circuits: ablating 93 non-consensus edges to zero causes cascading disruption, while the 7-edge CSS$=1.0$ core suffers minimal cascade loss. Under mean or resample ablation, which mitigate this size bias, the advantage is within noise.

SVA and Arithmetic show negative Δ under all ablation methods, confirming their failure is robust.

Implication. IOI's positive result requires the conjunction of ACDC `isinf` sparsity *and* zero ablation bias. Removing either confound substantially reduces or eliminates the advantage.

5 Cross-Model Scaling Failure

5.1 Qwen-2.5 0.5B: IOI

To test generalization, we apply CSS to Qwen-2.5 0.5B on IOI. The model has 179,388 total edges—5.5$\times$ larger than GPT-2 Small's 32,491.

At top-0.3% binarization ($\sim$538 edges per method), pairwise Jaccard values collapse: $J(\text{EAP-IG}, \text{EP}) = 0.042$, $J(\text{EAP-IG}, \text{ACDC}) = 0.006$, $J(\text{EP}, \text{ACDC}) = 0.008$. The CSS$=1.0$ core is **empty** (0 edges). Even the relaxed CSS$\geq 2/3$ set contains only 66 edges with CPR $= 0.031$, far below single-method baselines (best: EAP-IG CPR $= 0.055$).

IFR vs. ACDC as third method. We tested both Iterative Feature Removal (IFR) and ACDC top-k as the third method (Table 8). Both fail identically: IFR yields $J = 0.000$ with both gradient methods; ACDC yields $J \leq 0.008$. The failure is not method-specific but structural.

5.2 Qwen-2.5 0.5B: Arithmetic

The pattern replicates on Arithmetic: $J_{\max} = 0.038$, CSS$=1.0 = 0$ edges, CSS$\geq 2/3 = \sim 55$ edges with CPR $= 0.025$. Table 9 confirms that all Qwen tasks exhibit the same degenerate pattern.

5.3 Combinatorial Dilution

The mechanism is *combinatorial dilution*: when the edge space grows by $5.5\times$ but circuit structure does not scale linearly, a fixed top-k percentage selects non-overlapping subsets. At GPT-2 scale, top-0.3%

Task	Zero Δ	Mean Δ	Resample Δ
IOI	$+0.080$	$+0.007$	$+0.003$
SVA	-0.470	-0.002	-0.020
Arithmetic	-0.053	-0.066	-0.070

Table 6: CSS advantage ($\Delta = \text{CSS}_{1.0}$ CPR $-$ best single CPR) by ablation method. IOI's advantage shrinks from $+0.080$ (zero) to $+0.003$ (resample), entering the noise floor.

	GPT-2 Small	Qwen-2.5 0.5B	Ratio
Total edges	32,491	179,388	5.5×
Top-k (0.3%)	∼100	∼538	5.4×
Max Jaccard ($J_{\text{EAP-IG}\times\text{EP}}$)	0.307	0.042	7.3× ↓
CSS$=1.0$ edges	7	0	—
CSS$=1.0$ CPR	0.109	—	—

Table 7: GPT-2 vs. Qwen-2.5 edge space scaling. Jaccard drops superlinearly with edge space size, rendering consensus empty on the larger model.

yields ∼100 edges from a 32k pool; at Qwen scale, ∼538 edges from 179k. The IOI circuit distributes across more layers and heads in the larger model, and each method traces a different subset of the expanded computation paths.

Extrapolating the Jaccard decay ($0.307 \to 0.042$ over $5.5\times$ edge space growth), models at Gemma-2 2B scale (∼500k edges) would yield $J_{\max} \approx 0.01$, and LLaMA-3.1 8B (∼2M edges) would approach random intersection baselines ($J \approx 0.003$). We did not run these experiments, as the scaling trend is already clear.

6 Comparison with CIRCUS

CIRCUS [Parekh et al., 2026] constructs consensus *within* a single method by varying the pruning threshold. We compare both consensus approaches on GPT-2 tasks (Table 10).

CIRCUS produces negative CPR on all three tasks, performing worse than random ablation. Its within-method threshold sweep constructs nested sets (k=50 $\subset$ k=100 $\subset$ k=150) whose intersection equals the most conservative cut—it cannot leverage cross-method information. CSS outperforms CIRCUS on IOI where methods genuinely disagree, but neither method provides reliable consensus on the majority of tasks.

7 Jaccard as a Diagnostic

Given that CSS fails when methods are too similar (high Jaccard) or too different (low Jaccard at scale), we examine whether pairwise Jaccard can serve as a *diagnostic* for when consensus is viable.

Table 11 reveals that with only 3 valid data points, no statistically significant correlation exists (Spearman $\rho = -0.50$, $p = 0.67$). The sole positive data point (IOI×GPT-2) is confounded and serves as a Cook's $D > 5.0$ influential point that drives the entire negative trend. The Qwen data points are degenerate—CSS advantage is undefined when the consensus set is empty.

Rather than "Jaccard predicts consensus utility," the data suggest a **Goldilocks zone** hypothesis: consensus requires Jaccard high enough for non-empty intersection, yet low enough that the intersection carries information beyond any single method. Our experiments show that the upper bound (too-high Jaccard) manifests on GPT-2 SVA/Arithmetic, while the lower bound (too-low Jaccard) manifests on Qwen. The viable zone—if it exists—is narrow and task-specific.

Third method	$J(\cdot, \text{EAP-IG})$	$J(\cdot, \text{EP})$	CSS$= 1.0$
IFR	0.000	0.000	0 edges
ACDC	0.006	0.008	0 edges

Table 8: Qwen-2.5 IOI: both candidate third methods produce near-zero overlap with gradient methods. The failure is independent of method choice.

Task	CSS$= 1.0$ edges	$J_{\max}$	Verdict
IOI (IFR)	0	0.000	FAIL
IOI (ACDC)	0	0.042	FAIL
Arithmetic (ACDC)	0	$\sim$0.038	FAIL

Table 9: All Qwen-2.5 0.5B configurations fail with the same pattern: near-zero Jaccard and empty consensus core.

8 Discussion

Why does IOI succeed? IOI on GPT-2 is a well-studied task with a known ground-truth circuit [Wang et al., 2023]. The three methods discover genuinely different subsets of this circuit: EAP-IG and EP favor different attention heads, and ACDC's greedy search identifies a distinct sparse backbone. The $J_{\max} = 0.307$ reflects real algorithmic diversity, and the 7-edge consensus core captures high-confidence edges. However, even this success is fragile: under mean ablation, the advantage drops to $+0.007$.

When methods are not independent. EAP-IG and EP are both gradient-based and share the same underlying signal (gradient of the patching effect). On SVA, where number-tracking heads in layers 7–10 dominate, both methods converge on the same edges ($J = 0.653$). True K-method consensus requires methods from genuinely different algorithmic families—a condition that is harder to satisfy than it appears.

The scaling wall. The combinatorial dilution effect is perhaps the most significant finding. Edge-level consensus fundamentally assumes that different methods can "see" the same edges. In larger models, the same behavioral circuit may be implemented through different edge-level paths, and each method discovers a different valid decomposition. This is not a failure of any individual method but a structural property of edge-level analysis in deep networks.

Implications for mechanistic interpretability. Our results suggest that edge-level binary consensus is not a viable path to more reliable circuit identification, at least not without addressing the independence and scaling challenges. Possible directions include: (1) consensus at the *node* (attention head) rather than edge level, where combinatorial dilution is less severe; (2) soft consensus using calibrated scores rather than binary membership; and (3) consensus conditional on method family, explicitly accounting for inter-method correlation.

9 Related Work

Circuit discovery. Activation patching [Vig et al., 2020, Geiger et al., 2021] and its automated variants [Conmy et al., 2023, Syed et al., 2023, Hanna et al., 2024] identify circuits by measuring the causal effect of interventions. Wang et al. [2023] manually identified the IOI circuit in GPT-2.

Cross-method comparison. Mondorf and Plank [2025] compared gradient-based circuit discovery methods and proposed score averaging. Their work focuses on methods within the gradient family; we

Method	IOI CPR	SVA CPR	Arith CPR	Tasks +
CIRCUS	−0.108	−0.150	−0.120	0/3
CSS= 1.0	+0.109	−0.820	−0.184	1/3
Best single	+0.029	−0.348	−0.131	0/3

Table 10: Comparison with CIRCUS (zero ablation). Both consensus methods are severely limited. CIRCUS fails on all tasks; CSS succeeds only on IOI.

Task×Model	$J_{\max}$	CSS Δ	Quality	Confounds
IOI×GPT-2	0.307	+0.080	confounded	isinf + zero
SVA×GPT-2	0.653	−0.002	neutral	$\Delta \approx$ noise
Arith×GPT-2	0.695	−0.066	negative	—
IOI×Qwen	0.042	undef.	degenerate	CSS= 1.0 empty
Arith×Qwen	0.038	undef.	degenerate	CSS= 1.0 empty

Table 11: Jaccard–utility relationship across all task×model pairs. Only $n = 3$ valid points exist; $n = 2$ are degenerate.

extend to cross-family consensus. Parekh et al. [2026] proposed CIRCUS, a within-method consensus approach using threshold diversity.

Circuit faithfulness evaluation. Hanna et al. [2024] introduced systematic faithfulness metrics. Our ablation sensitivity analysis (§4.3) highlights that faithfulness measurements are sensitive to the choice of ablation method—a confound that affects all circuit evaluation, not just consensus approaches.

10 Limitations

Limited task and model coverage. We evaluate only 3 completed tasks on GPT-2 Small and 2 on Qwen-2.5 0.5B. The factual recall experiment was abandoned due to task–model mismatch (GPT-2 Small baseline accuracy $= 52\%$), and larger models (Gemma-2 2B, LLaMA-3.1 8B) were not attempted due to the established scaling trend and hardware constraints.

Method family diversity. Two of our three methods (EAP-IG, EP) are gradient-based. This limits the effective diversity to $K \approx 2$ in some settings. A truly diverse ensemble would require methods from more distinct algorithmic families.

Sample size for correlation analysis. With $n = 3$ valid data points, we cannot establish statistical relationships between Jaccard and consensus utility. The Goldilocks zone hypothesis remains speculative.

11 Conclusion

We present a systematic empirical analysis of cross-method circuit consensus, evaluating CSS across multiple tasks and model scales. Our central finding is negative: binary edge-level consensus fails in 4 out of 5 task–model configurations, and the sole positive case (IOI on GPT-2) is confounded by binarization artifacts and ablation method bias.

Three structural factors explain the failure: (1) insufficient method independence, particularly between gradient-based approaches; (2) binarization pathologies that create misleading diversity signals; and (3) combinatorial dilution that renders edge-level consensus impossible at scale.

These findings carry a constructive message: before investing in consensus-based circuit refinement, practitioners should verify that their methods produce genuinely independent evidence—measurable via

Analysis	ρ	p	n	Interpretation
All valid points	-0.50	0.67	3	not significant
Remove IOI (confounded)	undef.	—	2	cannot compute
Include Qwen ($\Delta{=}-0.05$ proxy)	-0.30	0.62	5	not significant

Table 12: Jaccard–utility correlation is not statistically significant under any analysis.

pairwise Jaccard—and that the edge space is small enough for meaningful overlap. The Goldilocks zone for viable consensus appears narrow, task-specific, and difficult to guarantee a priori.

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